

ASHEVILLE 13 S, NC, USA

WMO: 746970

Lat: 35.4185N

Lon: 82.5567W

Elev: 641

StdP: 93.86

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.6	-7.2	-16.4	1.0	-6.2	-14.4	1.2	-4.1	5.7	11.8	5.1	11.0	1.8	N/A	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	30.4	21.3	29.3	21.2	28.2	20.8	23.0	27.7	22.6	27.0	22.0	26.2	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.8	24.8	21.2	17.2	24.4	20.7	16.6	24.0	71.9	27.9	69.8	27.1	67.7	26.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.8	4.0	3.5	-13.4	32.3	2.5	1.4	-15.2	33.3	-16.7	34.1	-18.1	34.9	-19.9	36.0
			-14.0	24.2	2.5	0.7	-15.8	24.7	-17.3	25.1	-18.7	25.5	-20.5	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	2.7	4.5	8.4	13.1	17.2	21.2	22.6	22.2	19.2	13.5	8.0	4.7
	DBStd	8.02	5.45	4.81	4.72	3.91	3.38	2.10	1.86	1.83	2.87	4.16	4.34	4.67
	HDD10.0	777	232	162	88	16	1	0	0	0	0	15	88	174
	HDD18.3	2318	484	387	308	161	59	4	1	1	24	158	311	422
	CDD10.0	1928	6	8	39	107	226	335	391	379	276	123	28	11
	CDD18.3	429	0	0	0	3	26	89	132	122	51	7	1	0
	CDH23.3	2887	0	1	6	69	235	626	843	718	336	51	2	0
CDH26.7	632	0	0	0	5	32	140	216	169	63	6	0	0	
Wind	WSAvg	1.4	1.8	1.8	1.8	1.8	1.4	1.2	1.0	1.0	1.0	1.2	1.5	1.6
Precipitation	PrecAvg	1498	125	116	137	120	109	141	139	143	131	96	107	135
	PrecMax	2150	297	237	256	221	253	269	361	256	491	223	219	259
	PrecMin	1008	33	49	51	54	37	40	49	26	24	7	38	59
	PrecStd	257	63	46	49	41	54	56	69	57	90	57	46	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	20.9	24.1	27.4	29.4	31.3	31.7	31.7	30.2	27.0	23.0	19.7
		MCWB	13.0	14.3	14.5	16.7	19.0	20.9	22.2	21.6	19.7	19.9	15.0	13.8
	2%	DB	16.4	17.9	21.7	25.4	27.5	29.7	30.2	30.1	28.5	24.7	20.2	17.5
		MCWB	12.3	12.3	13.5	16.3	18.6	20.8	22.0	21.8	20.0	17.4	13.3	13.4
	5%	DB	14.3	15.8	19.5	23.6	26.0	28.4	29.1	28.6	27.0	22.6	18.2	15.1
		MCWB	10.4	11.0	12.5	15.0	18.1	20.6	21.8	21.6	19.8	16.7	13.3	11.8
	10%	DB	11.7	13.2	17.1	21.6	24.4	26.9	27.9	27.2	25.4	20.7	16.4	12.9
		MCWB	8.9	8.6	11.3	14.2	17.7	20.0	21.5	21.2	19.4	15.9	12.1	10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	15.8	16.1	19.9	21.0	23.0	23.9	24.0	22.3	21.2	18.1	16.8
		MCDB	17.6	19.1	21.3	23.2	26.2	27.9	29.0	28.4	26.4	25.1	19.6	18.4
	2%	WB	13.2	14.1	15.0	17.8	20.2	22.3	23.0	23.1	21.6	19.7	16.0	14.6
		MCDB	15.5	16.9	19.4	22.7	25.0	27.2	28.0	27.4	25.6	22.7	18.2	16.3
	5%	WB	11.3	11.8	13.9	16.6	19.4	21.6	22.6	22.6	20.8	18.5	14.8	12.6
		MCDB	13.1	14.3	17.6	21.0	23.9	26.4	27.3	26.6	24.5	20.9	17.2	14.2
	10%	WB	9.3	9.4	12.7	15.4	18.7	21.0	22.1	21.9	20.2	17.1	13.0	10.5
		MCDB	11.2	12.3	16.0	19.5	22.8	25.3	26.5	25.6	23.5	19.5	15.2	12.3
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.3	12.0	13.0	11.8	11.2	10.2	10.1	10.6	12.0	12.9	10.5
		MCDBR	15.1	15.8	15.9	16.8	14.6	13.3	12.1	12.3	13.3	14.2	15.7	13.1
		MCWBR	10.1	10.1	8.8	8.3	6.8	5.6	5.1	5.1	5.8	7.5	9.4	9.0
	5% WB	MCDBR	11.4	12.6	12.6	13.0	11.6	11.2	10.5	10.4	10.4	10.0	11.0	10.5
		MCWBR	8.9	9.5	8.0	7.3	5.9	5.1	4.7	4.7	5.0	5.8	7.9	8.6
Clear-Sky Solar Irradiance	taub	0.295	0.303	0.334	0.373	0.427	0.453	0.493	0.491	0.387	0.353	0.314	0.292	
	taud	2.543	2.511	2.454	2.343	2.209	2.170	2.085	2.072	2.349	2.439	2.514	2.579	
	Ebn at Noon	917	946	938	911	861	836	799	792	869	874	883	899	
	Edh at Noon	79	91	105	124	144	149	162	161	116	97	81	72	
All-Sky Solar Radiation	RadAvg	2.59	3.22	4.26	5.28	5.80	6.10	5.68	5.32	4.61	3.84	2.89	2.23	
	RadStd	0.16	0.35	0.27	0.40	0.43	0.42	0.39	0.25	0.41	0.40	0.25	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+142	+82
Regional (41 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+100

Nomenclature: See separate page